THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 66°	Their sum is 90°.
2	**(d)** 60°	The angles add up to 180° in 3 equal parts.
3	**(a)** 115°	Interior angles on the same side are supplementary.
4	**(c)** parallel	Use the tests for parallel lines.
5	**(b)** 28°	Vertically opposite angles are equal.
6	**(c)** acute	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.
7	**(a)** 90°	The angles around a point add up to 360°.
8	**(b)** 53°	Their sum is 90°.
9	**(d)** 36°	The angles add up to 180° in 5 equal parts.
10	**(a)** 100°	Interior angles on the same side are supplementary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The angles are y and $2y$, so $3y = 180$; $y = 60$ (1); The angles are 60° and 120° (1).	1+1
12	Compare 75° and 75° (1); Equal corresponding angles, so the lines are parallel (1).	1+1
13	Let the angle be y; then $y + (y + 20) = 180$; $2y = 160$ (1); $y = 80^\circ$ (1).	1+1
14	The angles are y and $3y$, so $4y = 180$; $y = 45$ (1); The angles are 45° and 135° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fourth angle $= 360^\circ - (95 + 85 + 110)^\circ = 70^\circ$ (1); Kinds: 95° obtuse, 85° acute, 110° obtuse, 70° acute (2).	1+2
16	For parallel lines these angles must be equal (1); Here 65° and 65° are equal (1); The lines are parallel (1).	1+1+1
17	The sum is 90° and the number of parts is 3 (1); One part $= 30^\circ$ (1); The angles are 30° and 60° (1).	1+1+1
	OR: Fourth angle $= 360^\circ - (120 + 70 + 60)^\circ = 110^\circ$ (1); Kinds: 120° obtuse, 70° acute, 60° acute, 110° obtuse (2).	1+2
18	For parallel lines these angles must be equal (1); Here 70° and 75° are not equal (1); The lines are not parallel (1).	1+1+1
	OR: The sum is 90° and the number of parts is 6 (1); One part $= 15^\circ$ (1); The angles are 15° and 75° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal to 50°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 130° (4 angles) (2); Check: $50 + 130 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $2y$ and $2y + y = 180$ (1); $y = 60$ (2); The angle is 120° and its supplement is 60° (2).	1+2+2
20	Equal to 110°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 70° (4 angles) (2); Check: $110 + 70 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $3y$ and $3y + y = 180$ (1); $y = 45$ (2); The angle is 135° and its supplement is 45° (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.

- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** parallel	Use the tests for parallel lines.
2	**(d)** 64°	Vertically opposite angles are equal.
3	**(a)** right	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.
4	**(d)** 110°	The angles around a point add up to 360°.
5	**(c)** 39°	Their sum is 90°.
6	**(c)** 30°	The angles add up to 180° in 6 equal parts.
7	**(b)** 68°	Interior angles on the same side are supplementary.
8	**(a)** not parallel	Use the tests for parallel lines.
9	**(d)** 97°	Vertically opposite angles are equal.
10	**(c)** obtuse	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare 110° and 105° (1); The corresponding angles are not equal, so the lines are not parallel (1).	1+1
12	Let the angle be y; then $y + (y + 40) = 180$; $2y = 140$ (1); $y = 70^\circ$ (1).	1+1
13	The angles are y and $4y$, so $5y = 180$; $y = 36$ (1); The angles are 36° and 144° (1).	1+1
14	Compare 62° and 62° (1); Equal corresponding angles, so the lines are parallel (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fourth angle $= 360^\circ - (45 + 150 + 100)^\circ = 65^\circ$ (1); Kinds: 45° acute, 150° obtuse, 100° obtuse, 65° acute (2).	1+2
16	For parallel lines these angles must be supplementary (1); Here 110° and 70° are supplementary (1); The lines are parallel (1).	1+1+1
17	The sum is 90° and the number of parts is 5 (1); One part $= 18^\circ$ (1); The angles are 36° and 54° (1).	1+1+1
	OR: Fourth angle $= 360^\circ - (80 + 130 + 40)^\circ = 110^\circ$ (1); Kinds: 80° acute, 130° obtuse, 40° acute, 110° obtuse (2).	1+2
18	For parallel lines these angles must be equal (1); Here 120° and 60° are not equal (1); The lines are not parallel (1).	1+1+1
	OR: The sum is 90° and the number of parts is 9 (1); One part $= 10^\circ$ (1); The angles are 40° and 50° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal to 65°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 115° (4 angles) (2); Check: $65 + 115 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $4y$ and $4y + y = 180$ (1); $y = 36$ (2); The angle is 144° and its supplement is 36° (2).	1+2+2
20	Equal to 130°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 50° (4 angles) (2); Check: $130 + 50 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $5y$ and $5y + y = 180$ (1); $y = 30$ (2); The angle is 150° and its supplement is 30° (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.

- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 70°	The angles around a point add up to 360°.
2	**(a)** 22°	Their sum is 90°.
3	**(d)** 20°	The angles add up to 180° in 9 equal parts.
4	**(b)** 45°	Interior angles on the same side are supplementary.
5	**(c)** 8	Use the tests for parallel lines.
6	**(a)** 143°	Vertically opposite angles are equal.
7	**(c)** reflex	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.
8	**(c)** 90°	The angles around a point add up to 360°.
9	**(a)** 7°	Their sum is 90°.
10	**(b)** acute	Classify by the measure: acute $< 90^\circ$, right $= 90^\circ$, obtuse between 90° and 180°, reflex above 180°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the angle be y; then $y + (y + 60) = 180$; $2y = 120$ (1); $y = 60^\circ$ (1).	1+1
12	The angles are y and $5y$, so $6y = 180$; $y = 30$ (1); The angles are 30° and 150° (1).	1+1
13	Compare 48° and 84° (1); The corresponding angles are not equal, so the lines are not parallel (1).	1+1
14	Let the angle be y; then $y + (y + 100) = 180$; $2y = 80$ (1); $y = 40^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fourth angle $= 360^\circ - (60 + 60 + 140)^\circ = 100^\circ$ (1); Kinds: 60° acute, 60° acute, 140° obtuse, 100° obtuse (2).	1+2
16	For parallel lines these angles must be equal (1); Here 98° and 98° are equal (1); The lines are parallel (1).	1+1+1
17	The sum is 90° and the number of parts is 9 (1); One part $= 10^\circ$ (1); The angles are 10° and 80° (1).	1+1+1
	OR: Fourth angle $= 360^\circ - (200 + 50 + 70)^\circ = 40^\circ$ (1); Kinds: 200° reflex, 50° acute, 70° acute, 40° acute (2).	1+2
18	For parallel lines these angles must be supplementary (1); Here 100° and 85° are not supplementary (1); The lines are not parallel (1).	1+1+1
	OR: The sum is 90° and the number of parts is 9 (1); One part $= 10^\circ$ (1); The angles are 70° and 20° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Equal to 75°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 105° (4 angles) (2); Check: $75 + 105 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $8y$ and $8y + y = 180$ (1); $y = 20$ (2); The angle is 160° and its supplement is 20° (2).	1+2+2
20	Equal to 145°: the angle itself, its vertically opposite angle, its corresponding angle and its alternate interior angle, that is 4 angles (1); Every other angle forms a linear pair with one of these, so it measures 35° (4 angles) (2); Check: $145 + 35 = 180$ (2).	1+2+2
	OR: Let the supplement be y; then the angle is $9y$ and $9y + y = 180$ (1); $y = 18$ (2); The angle is 162° and its supplement is 18° (2).	1+2+2

Mistakes to avoid:

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- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.

- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

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